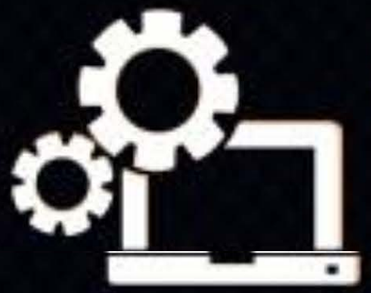
CS & IT ENGINEERING

DIGITAL LOGIC



Logic Gate

Lecture No. -6



By - CHANDAN SIR

Recap of Previous Lecture



Topic

Logic GATE

Topics to be Covered



- X-NOR GATE
- Questions
- Discussion

X-OR GATE :→

① 

② Truth table

A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

③ $A = B \quad y = 0$

$A = \bar{B} \quad y = 1$

④ $A \oplus A = 0$

$A \oplus 0 = A$

$A \oplus \bar{A} = 1$

$A \oplus 1 = \bar{A}$

⑤ $A \oplus A \oplus A \cdots \oplus A = \begin{cases} 0 & \text{Even no. of A} \\ A & \text{Odd no of A} \end{cases}$

⑥

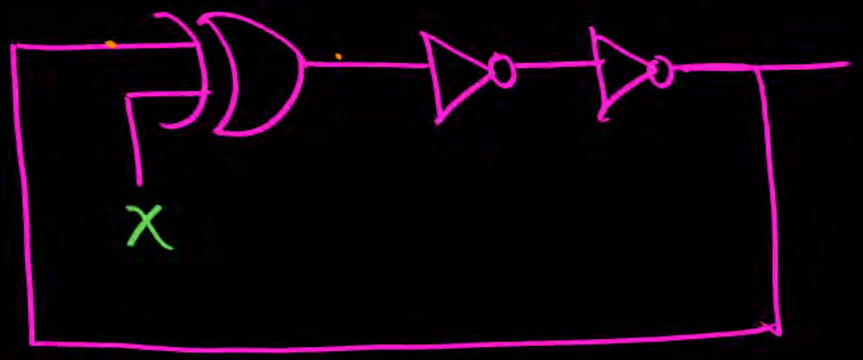


BUFFER



INVERTER

Q



Astable Multivibrator $X=1$

Bistable Multivibrator $X=0$

⑦ X-OR GATE follows commutative as well as associative law.

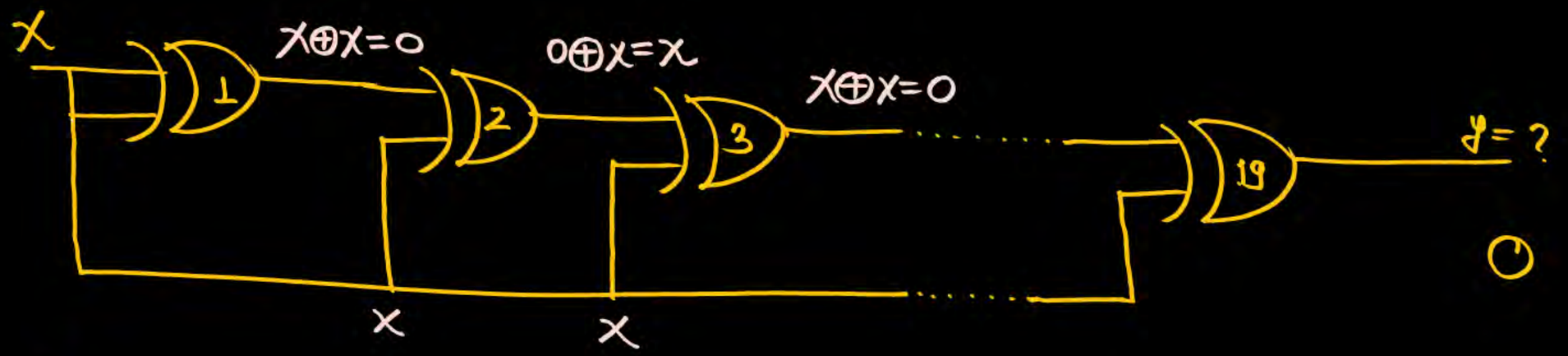
$A \oplus B = B \oplus A$

$(A \oplus B) \oplus C = A \oplus (B \oplus C)$

$A \oplus B \oplus C = \Sigma m(1,2,4,7)$

	A	B	C	$(A \oplus B) \oplus C$	$A \oplus (B \oplus C)$
0 →	0	0	0	0	0
✓1 →	0	0	1	1	1
✓2 →	0	1	0	1	1
3 →	0	1	1	0	0
✓4 →	1	0	0	1	1
5 →	1	0	1	0	0
6 →	1	1	0	0	0
✓7 →	1	1	1	1	1

Q



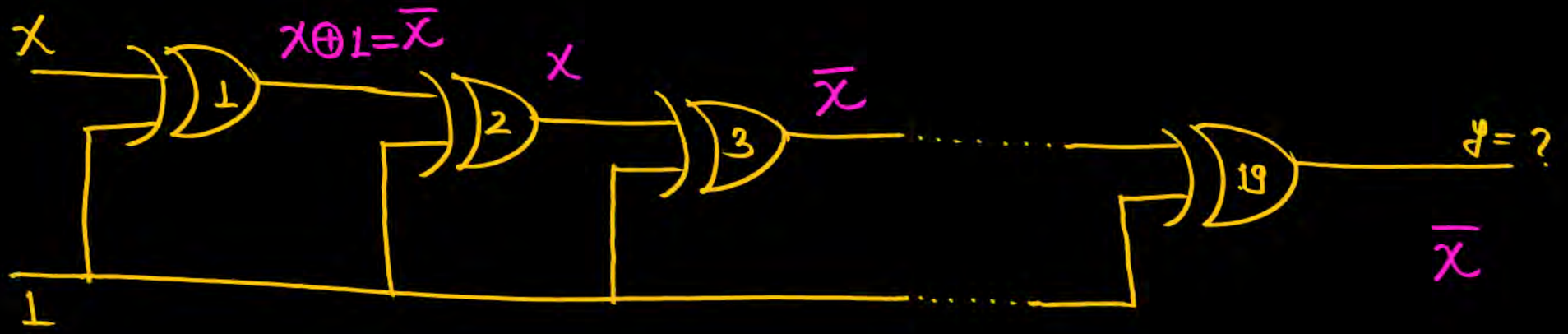
(A) 0

(B) X

(C) $\bar{X}$

(D) 1

Q



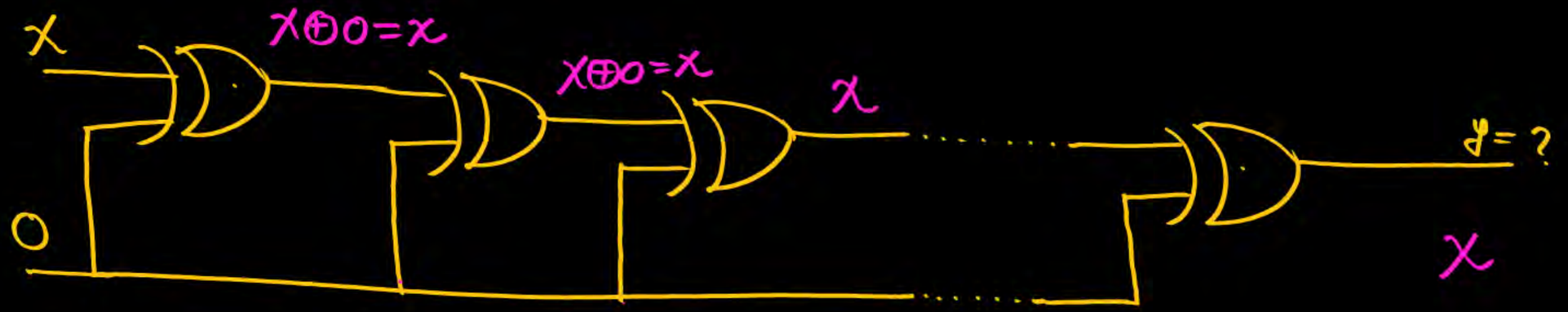
(A) 0

(B) X

~~(C) X̄~~

(D) 1

Q =



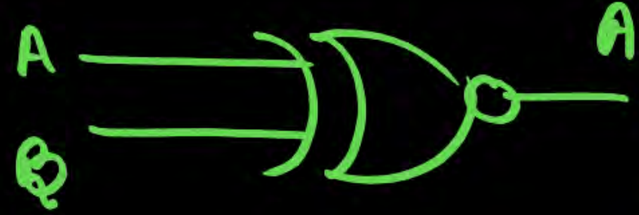
(A) 0

(B) X

(C) $\bar{X}$

(D) 1

Topic : X-NOR GATE



$$A \odot B = \overline{A \oplus B}$$

$$= \bar{A}\bar{B} + AB$$



Topic :X-NOR GATE

- **X-NOR GATE**

1. Symbol



2. Truth Table

A	B	y = A ⊙ B
0	0	1
0	1	0
1	0	0
1	1	1



Topic :X-NOR GATE

- **X-NOR GATE**

3. $A = B$ $\quad y = 1$

   $A \neq B$ $\quad y = 0$

A	B	$y = A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1



Topic : X-NOR GATE

- **X-NOR GATE**

4. ✓ Even parity detector

✓ Equivalence Logic

✓ Coincidence Logic

Parity

Even no. of '1' detector

A	B	$y = A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

⑤ $A \odot A = 1$

$A \odot 1 = A$

$A \odot \bar{A} = 0$

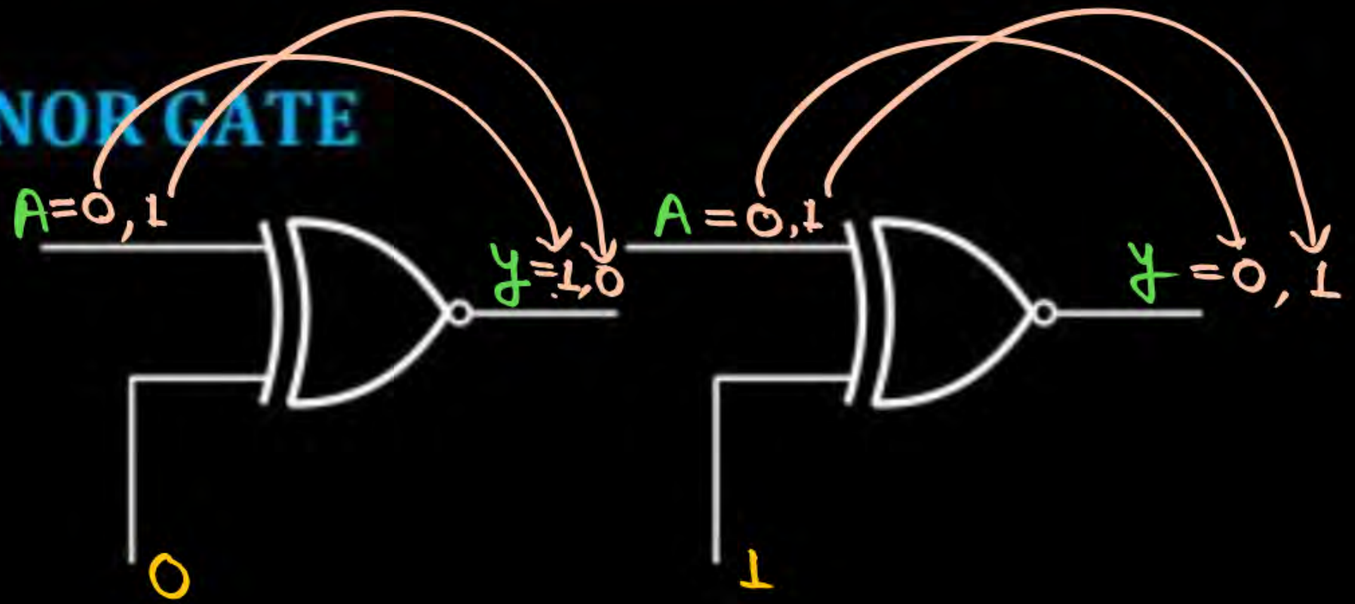
$A \odot 0 = \bar{A}$



Topic :X-NOR GATE

- **X-NOR GATE**

6.



INVERTER

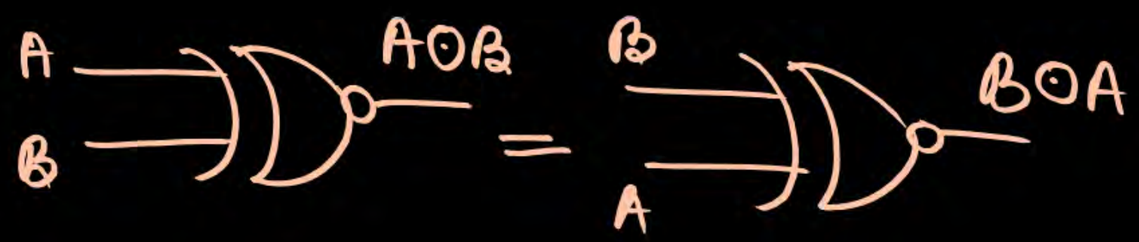
BUFFER



Topic : X-NOR GATE

- **Commutative Law**

$$A \odot B = B \odot A$$



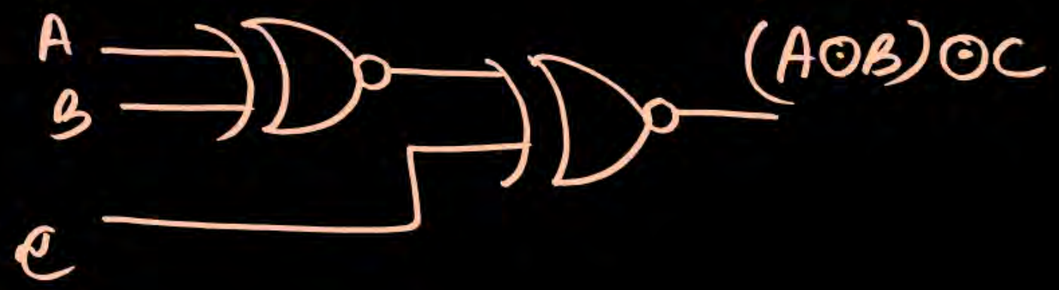


Topic :X-NOR GATE

- **Associate Law** ✓

2min

	A B C	A ⊙ (B ⊙ C)	(A ⊙ B) ⊙ C
0 →	0 0 0		0
1 →	0 0 1		1
2 →	0 1 0		1
3 →	0 1 1		0
4 →	1 0 0		1
5 →	1 0 1		0
6 →	1 1 0		0
7 →	1 1 1		1



$(A \odot B) \odot C = \Sigma m(1,2,4,7)$

$(A \odot B) \odot C = A \oplus B \oplus C$



Topic : X-NOR GATE

- For Even Number of Input

$$A \odot B = \overline{A \oplus B}$$

$$A \odot B \odot C \odot D = \overline{A \oplus B \oplus C \oplus D}$$



Topic :X-NOR GATE

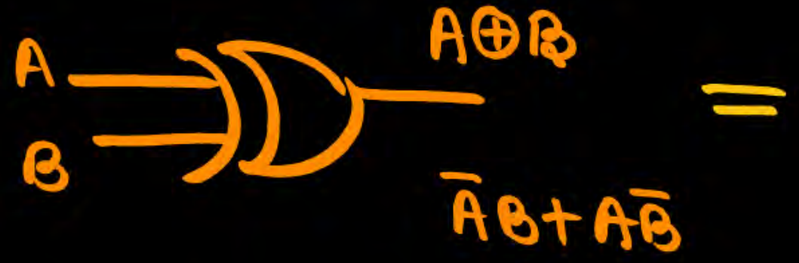
- **For Odd Number of Input**

$$(A \odot B) \odot C = A \oplus B \oplus C$$



Topic : X-NOR GATE

- **Alternate Symbol**



=



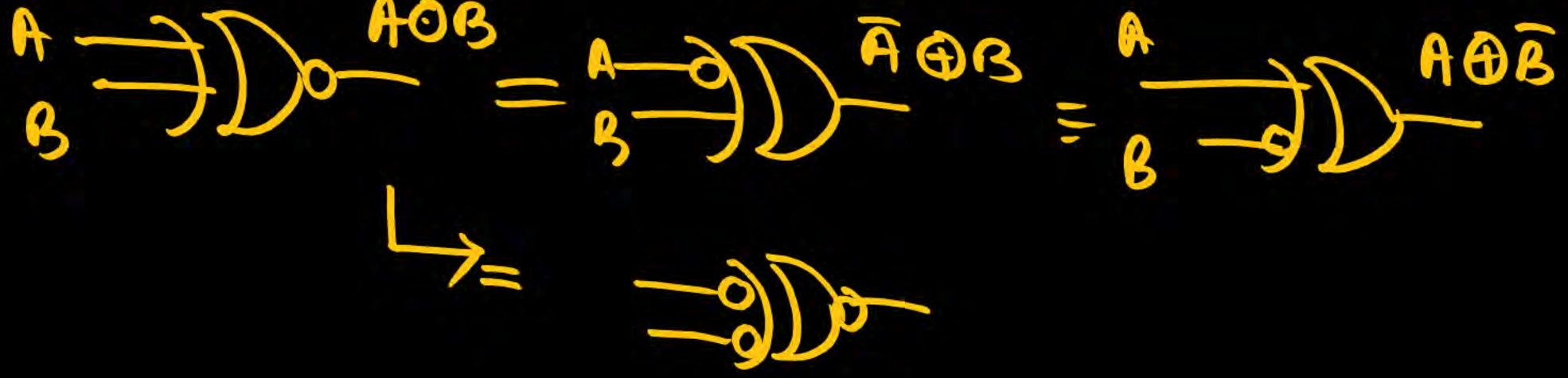
$A \odot \bar{B} = \bar{A}\bar{\bar{B}} + A\bar{B}$

$= \bar{A}B + A\bar{B}$

$= A \oplus B$

$$A \oplus B = \overline{A \odot B} = \bar{A} \odot B = A \odot \bar{B} = \bar{A} \oplus \bar{B}$$

$$A \odot B = \overline{A \oplus B} = \bar{A} \oplus B = A \oplus \bar{B} = \bar{A} \odot \bar{B}$$







#Q. The Boolean function given below: MSQ

HW. $f(A, B) = A \oplus B \oplus AB$

which statement is/are correct?

A. It is a OR GATE

B. It is a NAND GATE

C. It requires 3 NAND gate to implement the function

D. It required 2 NAND gates to implement the function

MCQ

#Q. Minimized expression will be $Y = A \oplus (A + B)$

HW

A $A \oplus B$

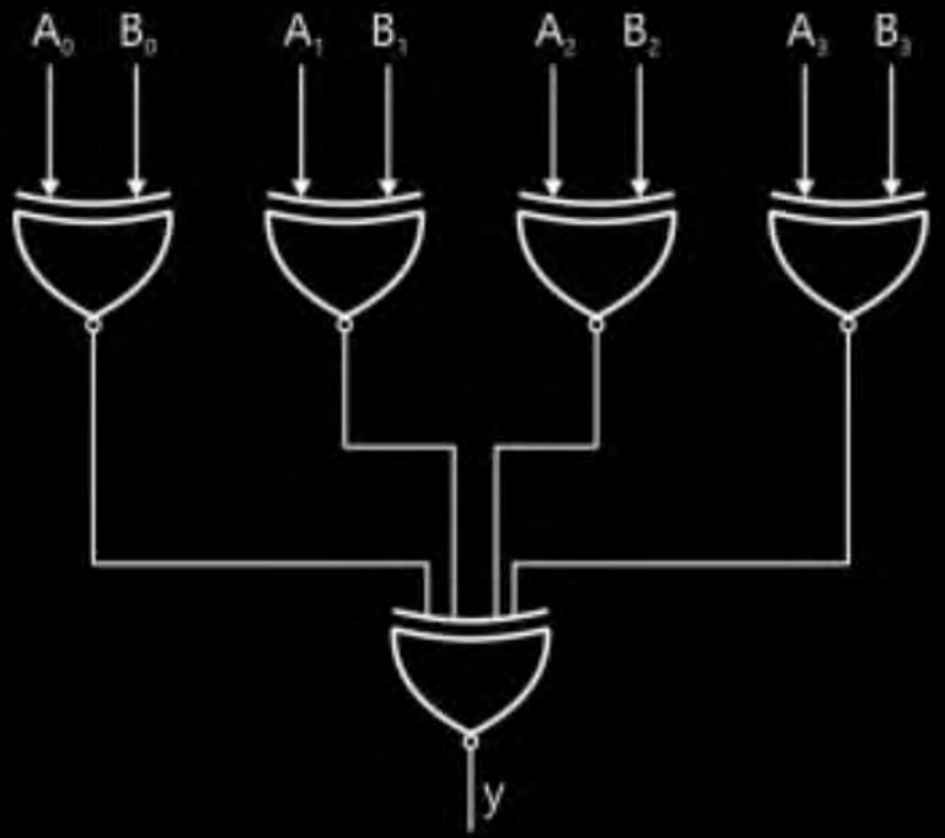
B $A \odot B$

C $\bar{A} \cdot B$

D $A + B$

#Q. If the output y = 1
Then correct input is/are-

HW



A 1111, 0000

B 1010, 0111

C 0101, 0101

D 1100, 1110

#Q. Output y will be-

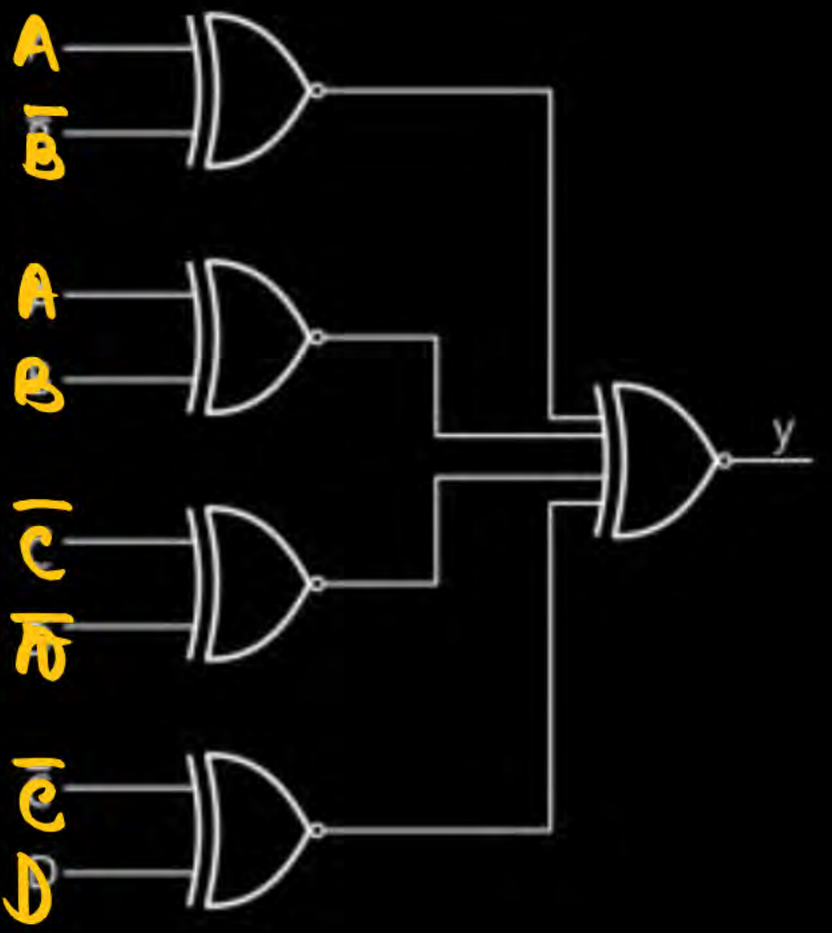
HW

(A) 0

(B) 1

(C) $A \oplus B$

(D) $A \oplus B \oplus C \oplus D$





2 Min Summary

X-NOR GATE

Questions

Discussion

Thank You GW Soldiers!

